Queen Mary
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EBU5376
Digital Signal Processing

Week 3 – Tutorial 1

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Tutorial Lecture 3.1 3.2

Covered in this tutorial:

- Types of DSP Systems
- Difference Equation
- Impulse Response
- Step Response

Types of DSP Systems

- Common types of DSP systems:

1. Linear systems
2. Time-invariant systems
3. Causal / Non-causal / Anti-causal systems
4. Memoryless systems
5. Stable / Unstable / Marginally Stable systems

A system may exhibit more than one properties at the same time

Difference Equation

- Definition: The difference equation expresses the outputs $y[n], y[n-1], \ldots$ of the system as a function of inputs $x[n], x[n-1], \ldots$.

- The general form of a difference equation for an LTI system can be represented as:

$$y[n] + a_1 y[n-1] + \cdots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + b_M x[n-M]$$

where $a_1, \ldots, a_N$ and $b_0, \ldots, b_M$ are the coefficients of the difference equation.

- M and N are the memory lengths for input $x[n]$ and output $y[n]$.

Difference Equation

- The general form can be reformulated as

$$y[n] = -a_1 y[n-1] - \cdots - a_N y[n-N] + b_0 x[n] + b_1 x[n-1] + \cdots + b_M x[n-M]$$

Current **output**

Past **outputs**

Current and past **inputs**

- Equivalently,

$$y[n] = -\sum_{i=1}^{N} a_i y[n-i] + \sum_{j=0}^{M} b_j x[n-j]$$

*This is a **recursive** equation.*

Solving Difference Equation with Initial Conditions

- Steps to solve a difference equation via z-transform:
  1. Reformulate the difference equation, with the input and the output on separate sides of the equation
  2. Take the z-transform on both sides
  3. Substitute $Y[z] = Z(y[n]) = y[0] + z^{-1}y[1] + z^{-2}y[2] + \cdots$
     - Use the initial condition to simplify the equation
  4. Apply partial fraction decomposition
  5. Perform inverse z-transform to obtain the solution $y[n] = \cdots$

Impulse Response, Step Response

- If the difference equation is *recursive*,
  - The **impulse response** $h[n]$ is the solution to the difference equation, when the input is the unit impulse function $\delta[n]$
    - It provides a system's natural response to a sudden, unit magnitude input at time zero.
  - The **step response** $s[n]$ is the solution to the difference equation, when the input is the unit step function $u[n]$
    - It shows how the system responds to a sustained change in input.

Question 1, Part 1

- Identify the linear systems from the options below.

  a. $y[n] = x[3n - 2]$

  b. $y[n] = x[n] + 2x[n - 1]$

  c. $y[n] = x[n - 2] + 3$

  d. $y[n] = x[-n]$

Question 1, Part 2

- Identify the time-invariant systems from the options below.

  a. $y[n] = x[3n - 2]$

  b. $y[n] = x[n] + 2x[n - 1]$

  c. $y[n] = x[n - 2] + 3$

  d. $y[n] = x[-n]$

Question 1, Part 3

- Identify the memoryless systems from the options below.

  a. $y[n] = x[3n - 2]$

  b. $y[n] = x[n] + 2x[n - 1]$

  c. $y[n] = x[n - 2] + 3$

  d. $y[n] = x[-n]$

Question 1, Part 4

- Identify the linear time-invariant (LTI) systems from the options below.

  a. $y[n] = x[3n - 2]$

  b. $y[n] = x[n] + 2x[n - 1]$

  c. $y[n] = x[n - 2] + 3$

  d. $y[n] = x[-n]$

Question 2

- Derive the **impulse response** of the following system

$$y[n] = 0.16y[n-2] + x[n], \qquad y[-1] = 0, y[-2] = 0$$

Question 3

- Derive the **step response** of the following system

$$y[n] = 0.5y[n-1] + x[n], \qquad y[-1] = 1$$

Question 4

- In this question, you'll be presented with various practical scenarios requiring analysis of DSP system responses.

- Decide whether to use **impulse or step response** for each situation, and briefly explain why it's the preferred choice

Question 4

- **Scenario 1**: Designing a noise-cancellation headphone system requires swift reaction to various sound levels. Which time-domain response would guide us in verifying system reactivity?

Question 4

- **Scenario 2**: A smart home thermostat is tested for its response to user-adjusted temperature settings. To ensure comfort, which time-domain response would best reflect the system's ability to reach and hold a new temperature?

Question 4

- **Scenario 3**: When testing for the airbag's deployment timing and precision, which time-domain response is essential to consider?

Question 5

- You're building an audio compressor that processes input samples to reduce sudden loud peaks. You consider the following system:

$$y[n] = \max(x[n], x[n-1])$$

Which of the following properties does this satisfy?

- Linearity
- Time-invariance
- Causality
- Memory

Briefly justify your answer.

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